Circuit, packet, and message switching

1. Datagram and
2. virtual circuit packet switching

Communication can be established in two ways between two or more devices that are connection-oriented and connection-less. Network layers can offer these two different types of services to its predecessor layer for transferring data. **Connection-oriented services** involve the establishment and termination of the connection while **connection-less services** don't require any connection creation and termination processes for transferring data.

Another difference between connection-oriented and connection-less services is connection-oriented communication uses a stream of data and is vulnerable to router failure while connection-less communication uses messages and is robust to router failure.

Definition of Connection-oriented Service

Connection-oriented service is analogous to the **telephone system** that requires communication entities to establish a connection before sending data. TCP provides Connection-oriented services as does **ATM, Frame Relay** and **MPLS** hardware. It uses **handshake process** to establish the connection between the sender and receiver. **Reliability** is achieved by having recipient acknowledge each message. There are **sequencing** and **flow control,** that's the reason packets received at the receiving end are always in **order**. It uses **circuit switching** for transmission of data.

Definition of Connection-less Service

Connection-less service is analogous to the **postal system**. In which packets of data (usually known as a **datagram**) is transmitted from source to destination directly. Each packet is treated as an individual entity, which allows communication entities to send data before establishing communication. Each packet carries a **destination address** to identify the intended recipient. Packets don't follow a **fixed path** that is the reason the packets received at receiver end can be out of order. It uses **packet switching** for transmission of data.

Comparison Chart

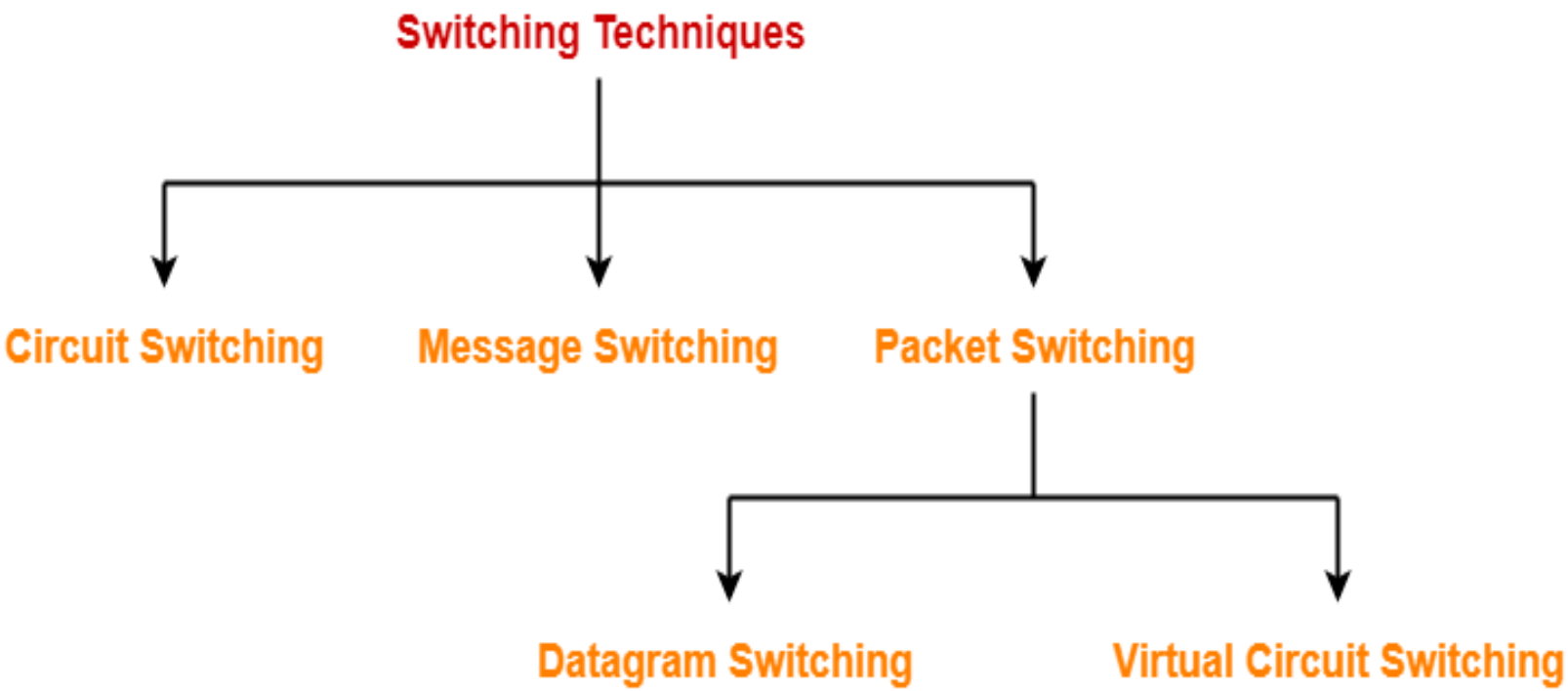
BASIS OF COMPARISON	CONNECTION-ORIENTED SERVICE	CONNECTION-LESS SERVICE
Prior Connection Requirement	Necessary	Not required
Reliability	Ensures reliable transfer of data.	Not guaranteed.
Congestion	Unlikely	Occur likely.
Transferring mode	It can be implemented using circuit switching and virtual circuit.	It is implemented using packet switching.
Lost data retransmission	Feasible	Practically, not possible.

BASIS OF COMPARISON	CONNECTION-ORIENTED SERVICE	CONNECTION-LESS SERVICE
Suitability	Suitable for long and steady communication.	Suitable for bursty Transmission.
Signalling	Used for connection establishment.	There is no concept of signalling.
Packet forwarding	Packets sequentially travel to their destination node and follows the same route.	Packets reach the destination randomly without following the same route.
Delay	There is a delay in transfer of information, but once the connection is established faster delivery can be achieved.	Because to the absence of connection establishment phase, the transmission is faster.
Resource Allocation	Need to be allocated.	No prior allocation of the resource is required.

Key Differences Between Connection-oriented and Connection-less Services

The points given below explains the difference between connection-oriented and connection-less services:

1. There is a requirement for prior connection for communication in connection-oriented services, in contrast, it is not needed in connection-less services.
2. Reliability is more in connection-oriented as compared to connection-less services.
3. Traffic congestion is greater in connection-less services whereas its occurrence is rare in connection-oriented services.
4. In connection-oriented services order of packets received at the destination is same as sent from the source. On the contrary, the order might change in connection-less services.
5. All packets follow the same path in connection-oriented services while packets follow a random path to reach the destination in connection-less services.
6. Connection-oriented service is appropriate for long and steady communication whereas connection-less service is fit for bursty transmission.
7. In connection-oriented services, sender and receiver are synchronized with each other while it is not the case of connection-less services.
8. Connection-oriented services use circuit switching on the other hand packet switching is used in connection-less services.
9. Bandwidth requirement is higher in Connection-oriented services whereas its low in connection-less services.


Switching Techniques
Circuit Switching
Message Switching
Packet Switching
Datagram Switching
Virtual Circuit Switching

Circuit Switching-

This switching technique operates in the following three phases-

1. Establishing a circuit
2. Transferring the data
3. Disconnecting the circuit

1. Establishing A Circuit-

In this phase,

- A circuit is established between the two ends.
- Circuit provides a dedicated path for data to travel from one to the other end.
- Resources are reserved at intermediate switches which are used during the transmission.
- The intermediate switches are connected by the physical links.

2. Transferring The Data-

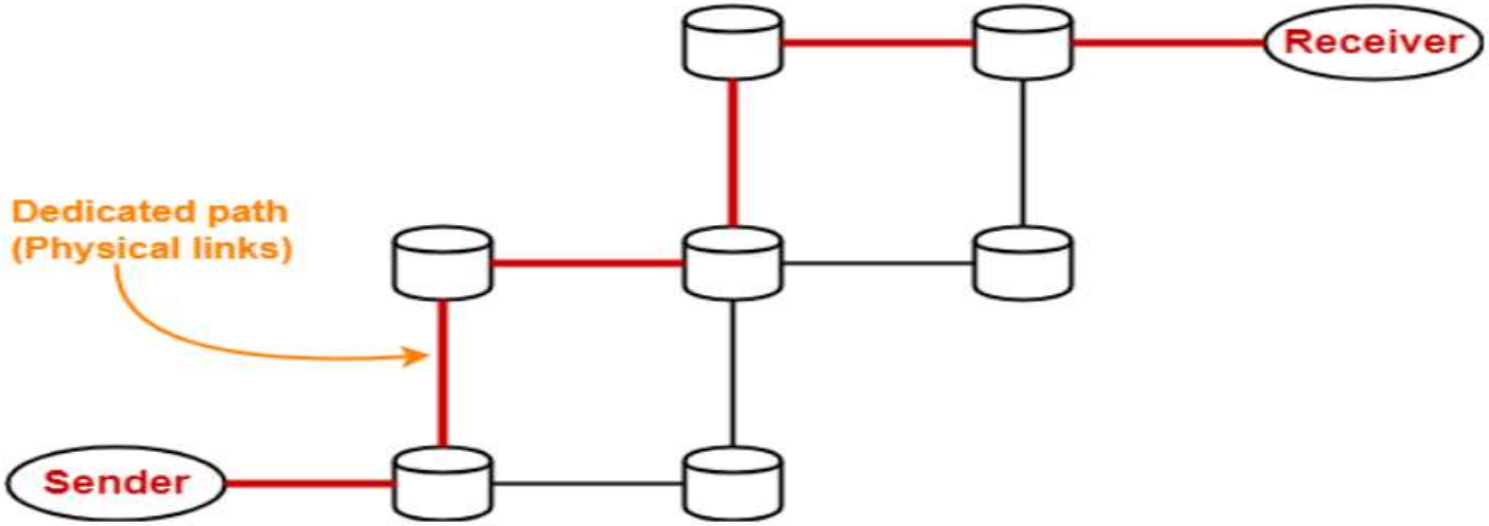
After the circuit is established,

- The entire data travels over the dedicated path from one end to the other end.

3. Disconnecting The Circuit-

After the data transfer is completed,

- The circuit is torn down i.e. disconnected.



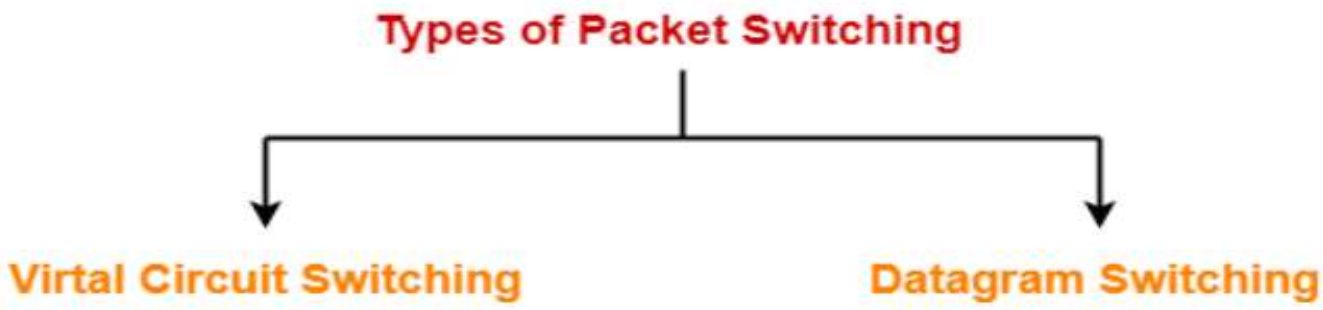
Message Switching-

In message switching,

- There exists no dedicated path to transfer data.
- The entire message is treated as a single data unit.
- The message is then forwarded from hop to hop.
- **Store and Forward** is an important characteristic of message switching.
- The message carries a header that contains the full information about the destination.
- When any intermediate switch receives the message, it stores the entire message.
- The message is stored until sufficient resources become available to transfer it to the next switch.
- When resources become available, the switch forwards the message to the next switch.

Types of Packet Switching-

Packet switching may be carried out in the following 2 ways-



- The entire message to be sent is divided into multiple smaller size packets.
- This process of dividing a single message into smaller size packets is called as **packetization**.
- These smaller packets are sent after the other.
- It gives the advantage of pipelining and reduces the total time taken to transmit the message.

Virtual Circuit Switching	Datagram Switching
The first packet during its transmission- 1) Informs the intermediate switches that more packets are following. 2) Reserve resources (CPU, bandwidth and buffer) for the following packets at all the switches on the way.	The first packet does not perform any such task during its transmission.
The packets are never discarded at intermediate switches and immediately forwarded since resources are reserved for them.	The packets may be discarded at intermediate switches if sufficient resources are not available to process the packets.
It is a connection oriented service since resources are reserved for the packets at intermediate switches.	It is a connection less service since no resources are reserved for the packets.
All the packets follow the same dedicated path.	All the packets take path independently.
Data appears in order at the destination since all the packets take the same dedicated path.	Data may appear out of order at the destination since the packets take path independently.
It is highly reliable since no packets are discarded.	It is not reliable since packets may be discarded.

Only first packet requires a global header which identifies the path from one end to other end. All the following packets require a local header which identifies the path from hop to hop.	All the packets require a global header which contains full information about the destination.
ATM (Asynchronous Transfer Mode) uses virtual circuit switching.	IP Networks use datagram switching.
Virtual circuit switching is normally implemented at data link layer.	Datagram switching is normally implemented at network layer.

Feature	Circuit Switching	Packet Switching
Dedicated Path	Yes	No
Path Formation	Path dedicated for one conversation	Route is established on a per packet basis of the conversation using datagram (or per conversation with virtual circuit)
Delay	Call setup delay	Packet transmission delay (call setup delay for virtual circuit)
Bandwidth Type	Fixed Bandwidth	Dynamic bandwidth
Overload Effects	Stops call establishment	Increases packet delay (can block call establishment and increase packet delay with virtual circuit)

Comparison of Virtual-Circuits & Datagrams

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

BASIS FOR COMPARISON	CIRCUIT SWITCHING	PACKET SWITCHING
Orientation	Connection oriented.	Connectionless.
Purpose	Initially designed for Voice communication.	Initially designed for Data Transmission.
Flexibility	Inflexible, because once a path is set all parts of a transmission follows the same path.	Flexible, because a route is created for each packet to travel to the destination.
Order	Message is received in the order, sent from the source.	Packets of a message are received out of order and assembled at the destination.

Technology/Approach	Circuit switching can be achieved using two technologies, either Space Division Switching or Time-Division Switching.	Packet Switching has two approaches Datagram Approach and Virtual Circuit Approach.
Layers	Circuit Switching is implemented at Physical Layer.	Packet Switching is implemented at Network Layer.

Key Differences Between Circuit Switching and Packet Switching

1. Circuit Switching is connection oriented that means a path is established between source and destination before the transmission occurs. On the other hand, Packet Switching is Connectionless that means a dynamic route is decided for each packet while transmission.
2. Circuit Switching was originally designed for voice communication whereas, Packet Switching was originally designed for data communication.
3. Circuit Switching is inflexible as once a path is established for transmission, it doesn't change while the duration of the session. On the other hand, Packet Switching is flexible as each packet may travel through a different route to reach its destination.
4. In packet switching, as each packet travels a different path hence, the packet are received out of order at the receiver side and later arranged in order. On the other hand, in circuit switching the entire message is received as it is as sent from a sender to receiver.
5. Space Division Switching or Time-Division Switching can be used to implement Circuit Switching whereas, Packet Switching can be implemented using two approaches Datagram Approach and Virtual Circuit Approach.
6. Circuit Switching is always implemented at physical layer whereas, Packet Switching is implemented on the network layer.

BASIS FOR COMPARISON	CIRCUIT SWITCHING	MESSAGE SWITCHING
Basic	A physical path is set between the two systems to place a telephone call.	The transfer of data is done by storing and forwarding it from and to the switching device.
Packet storage	Not stored directly sent.	The data is first stored then forwarded.
Transmission media	Analog and digital over various media	Digital over various media
Addressing	Geographical	Hierarchical
Routing	Manual type	Route is selected during call setup
Dedicated physical path	Not required	Necessary for the transmission
Cost	Greater than the message switching.	Reduced by using store and forward mechanism.

Key Differences Between Frame Relay and ATM

1. The packet size in the frame relay varies while ATM uses a fixed size packet known as a cell.
2. ATM produces fewer overheads as compared to the frame relay technology.
3. Frame relay is less expensive respective to the ATM.
4. ATM is faster than the frame relay.
5. ATM provides error and flow control mechanism, whereas the frame relay does not provide it.
6. Frame relay is less reliable than the ATM.
7. Throughput generated by frame relay is medium. In contrast, ATM has a higher throughput.
8. The delay in the frame relay is more. As against, it is less in case of ATM.

Advantages of the Frame Relay

Efficient communication process.

It performs fewer functions at the user-network interface.

Delay is also lowered.

Produces higher throughput.

It is cost effective.

It is faster than its predecessor X.25.

Advantages of the ATM

It can easily interface with the existing network such as PSTN, ISDN. It can be used over SONET/SDH.

Seamless integration with the different types of networks (LAN, MAN and WAN).

Effective utilization of the network resources.

It is less susceptible to the noise degradation.

Provides large bandwidth.

Disadvantages of the Frame Relay

Unreliable service.

The order of the arriving packets may not be maintained.

The erroneous packets are directly dropped.

The frame relay does not offer any flow control.

There is no provision of the acknowledgement of the received packets and retransmission control for the frames.

Disadvantages of the ATM

Cost of switching devices is higher.

Overhead generated by the cell header is more.

ATM QoS mechanism is quite complex.

BASIS FOR COMPARISON	FRAME RELAY	ATM
Packet size	Variable	Fixed
Processing overhead	Increased	Decreased
Data transfer	Implemented in more than one area network.	Takes place within a LAN
Cost	Inexpensive	Cost is higher
Speed	Low	High
QoS	Quantifiable QoS is not provided.	Offers quantifiable QoS.
Error control	No support is provided for error and flow control	Error and flow control is provided.
Data rate	64 Kbps up to 45 Mbps.	155.5 Mbps or 622 Mbps.
Reliability	Low	Good
Throughput	Medium	High
Delay	High	Less

Data Communication and Networking

Switching → Process of fwdg. Packets coming from one port to another port.

Switching → Connectionless / Connection-oriented.

Switching Methods

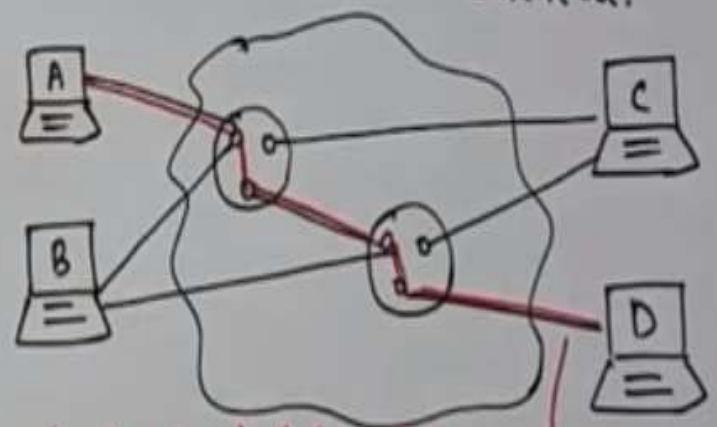
Switching Sends data along different routes.

(i) **CIRCUIT SWITCHING**:- It is a transmission mode that involves setting up a dedicated end to end Conn^n.

- ↳ Commonly used in Telephone Systems.
- ↳ Connection oriented.
- ↳ No delay in data flow
- ↳ Link of the Conn^n Can't be used to send any other data even when free.
- ↳ More bandwidth is required.
- ↳ Conn^n establish time is more.

Three Phases:-

↳ i) Circuit establishment
ii) Data Transfer
iii) Circuit disconnection.



A wants to send data to D.

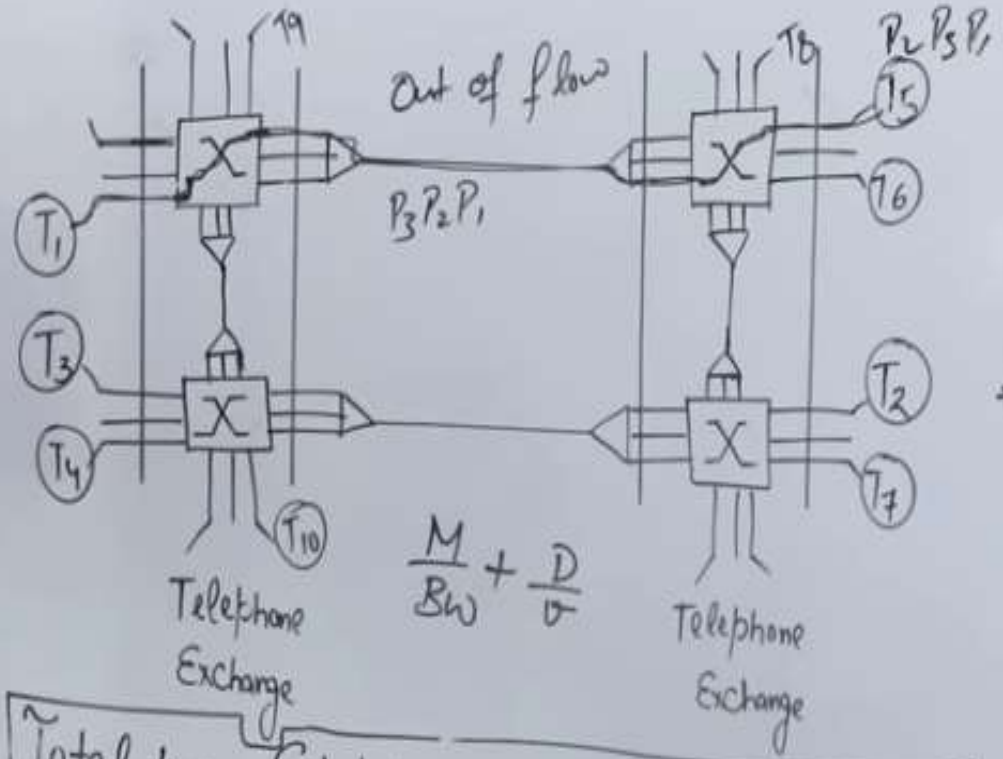
→ Route for A→D.

Circuit Switching.

Circuit Switching

CN

Dedicated Path



$$\frac{M}{Bw} + \frac{D}{v}$$

- Physical layer
- Contiguous flow
- No Readers
- Efficiency less
- Delays less

Total time = Setup time + TT + PD + Tear Down time

Data Communication and Networking

Switching Methods:

(ii) MESSAGE SWITCHING :- There is no dedicated path b/w two Communicating devices.

↳ Each Msg is treated as an independent unit and includes its own destination and Source addr.

→ Also Called as Store and Forward N/W. →

→ Efficient traffic management.

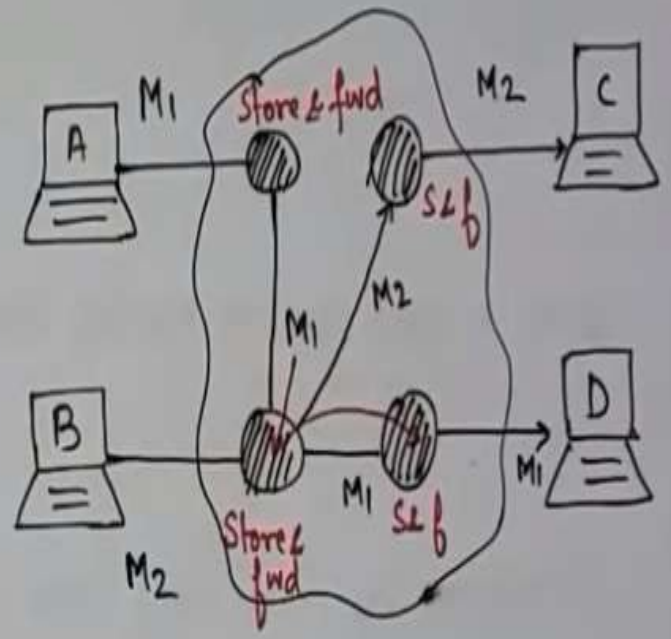
→ Reduces N/W traffic Congestion.

→ Some delay due to Storing and forwarding.

→ Large Storing Capacity is required.

↳ Used Commonly in Telegraph N/W

M1 - 101001



Message Switching.

"Message Switching"

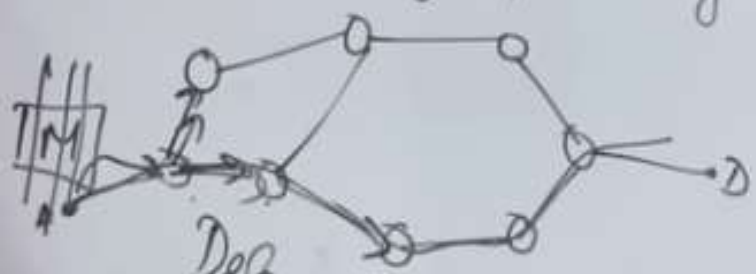
→ Predecessor of Packet Switching

→ Store and forward

→ Hop by Hop delivery

Hard disk

CS
↓
MS - 1960
↓
PS



Delay ↑ Eff ↑

Data Communication and Networking

Switching Method:

(iii) **PACKET SWITCHING:-** In this Messages are broken up into packets.

↳ Individual Packets take different routes to reach the destination.

↳ Advantages:-
- (i) Required Bandwidth ↓
- (ii) In Link failure diff. route can be chosen for remg. packets.

Datagram Packet Switching:- Message is divided into Stream of Packets.

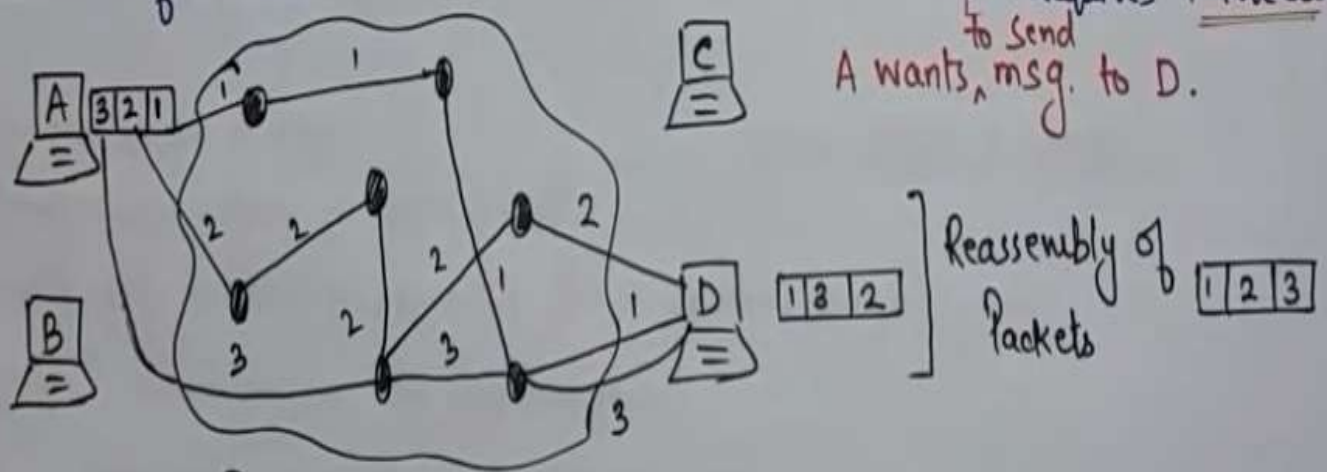
Virtual Circuit Packet Switching:-

Logical Connection is establish b/w the Sending and receiving devices.

↳ All the packets travel through the logical Conn^n → Virtual Circuit.

- → ↑ the bandwidth
- → ↓ transmission delay
- → Large amount of RAM is required.
- → Requires ↑ Processing Power.

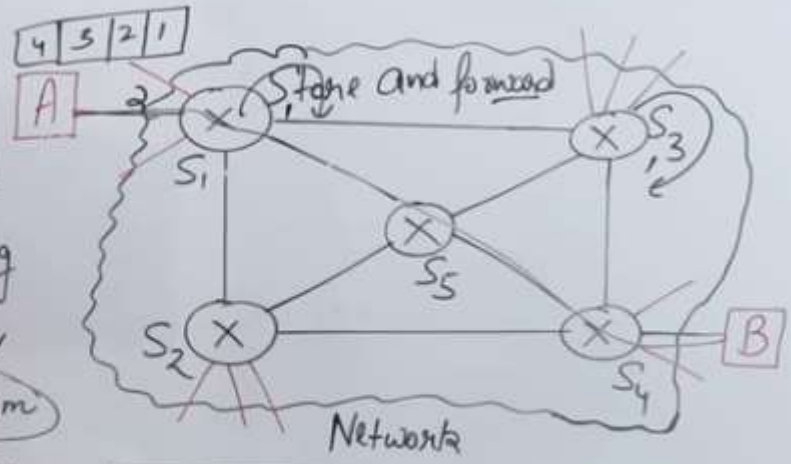
A wants to send msg. to D.



Packet Switching

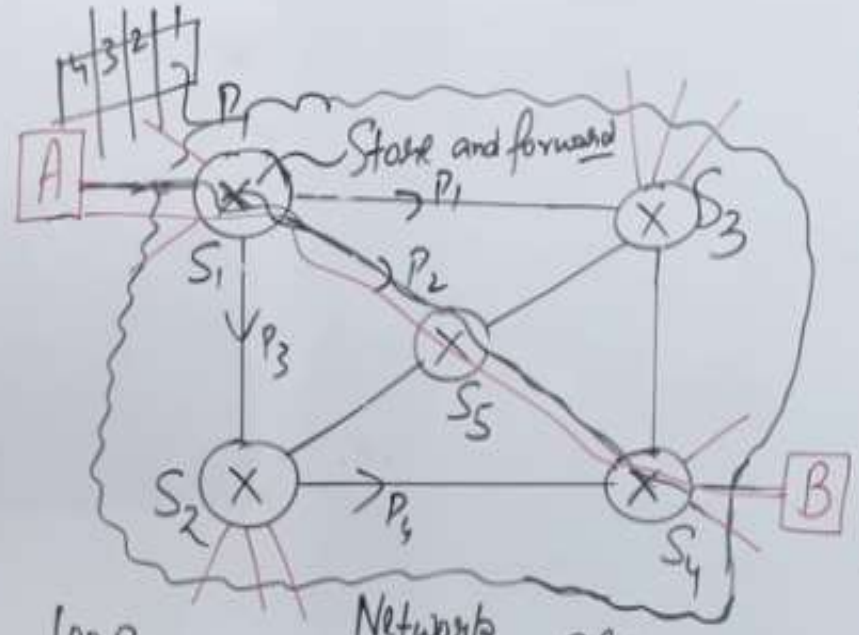
Packet Switching

- Datagram
- Virtual Circuit



$$\text{Total time} = n(TT) + PD$$

1) Data link and Network layer
2) Store and forward
3) Pipelining used
4) Efficiency ↑
5) Delay ↑



Network

Routing table

Cost

Delay

100B

Header ↑ 10B

global headers

IP

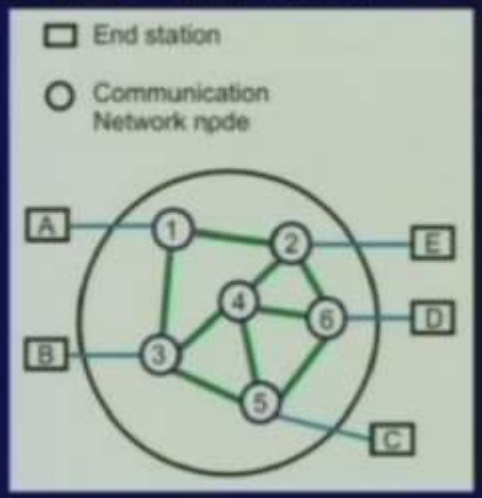
Datagram Switching	Virtual Circuit
* Connectionless	* Connection Oriented
* No Reservation	* Reservations
* Out of order	* Same order
* High overhead	* Less overhead
* Packet lost ↑	* Packet Lost ↓
* Used in Internet	* X.25, ATM
Eff ↑ Cost ↓	Cost ↑

- How two devices perform communication when there are many devices?
- One alternative is to establish point-to-point communication between each pair of devices using mesh topology
- Mesh topology is impractical for large number of devices
- A better alternative is to use switching techniques leading to switched communication network

Switched Communication Networks

- The end devices that wish to communicate are called stations. The switching devices are called nodes. Some Nodes connect to other nodes and some to attached stations.
- Network Topology not Regular
- Uses FDM or TDM for node to node communication
- There exists multiple paths between a source-destination pair for better network reliability.
- The switching nodes are not concerned with the contents of data.
- Their purpose is to provide a switching facility that will move data from node to node until they reach the destination.

Switched Communication Networks



Switching Techniques

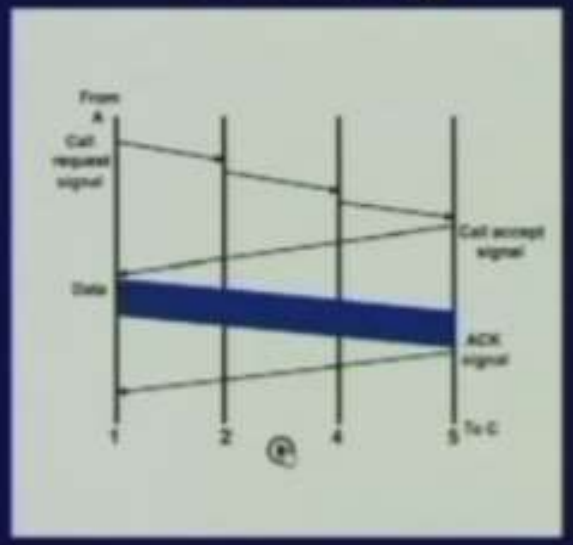
- Possible Switching techniques:
  - Circuit Switching
  - Message Switching
  - Packet Switching
- Communication via circuit switching implies that there is a dedicated communication path between two stations.
  - The path is a connected sequence of links between network nodes.
  - On each physical link, a logical channel is dedicated to the connection.

Circuit Switching

- Circuit establishment
  - To establish an end-to-end connection before any transfer of data.
  - Some segments of the circuit may be a dedicated link, while some other segments may be shared.
- Data transfer
  - Transfer data is from the source to the destination.
  - The data may be analog or digital, depending on the nature of the network.
  - The connection is generally full-duplex.
- Circuit disconnect
  - Terminate connection at the end of data transfer.
  - Signals must be propagated to deallocate the dedicated resources.

Circuit Switching

- ✓ Circuit Establishment
- ✓ Data Transfer
- ✓ Circuit Disconnect



Circuit Switching

- Originally developed for handling voice traffic, but is now also used for data traffic.
  - Once the circuit is established, the network is transparent to the users.
  - Information is transmitted at a fixed rate with no delay other than that required for propagation through the communication medium.
  - Best known example is the Public Switched Telephone Network (PSTN).

Circuit Switching

- Advantages:
  - Fixed bandwidth, guaranteed capacity (no congestion)
  - Low variance in end-to-end delay (constant delay)

Circuit Switching

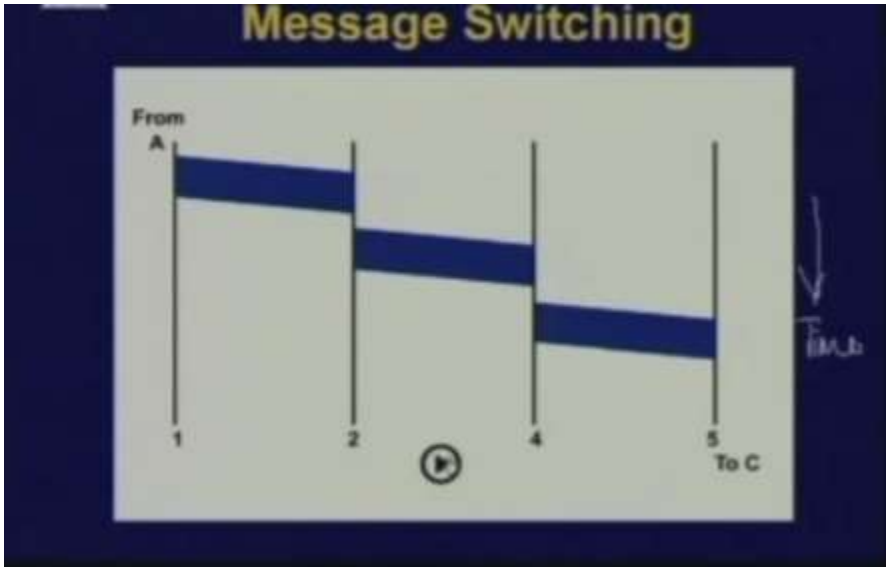
- Disadvantages:
  - Circuit establishment and circuit disconnect introduces extra overhead and delay.
  - Constant data rate from source to destination.
  - Channel capacity is dedicated for the duration of the connection, even if no data is being transferred.
  - For voice connection, utilization is typically high (Statistics: 64-73% time one speaker speaking, 3-7% time both are speaking, 20-33% time both are silent)
  - Inefficient for bursty data traffic. In a typical user/host data connection, line utilization is poor.
  - Other users cannot use it even it is free of traffic.

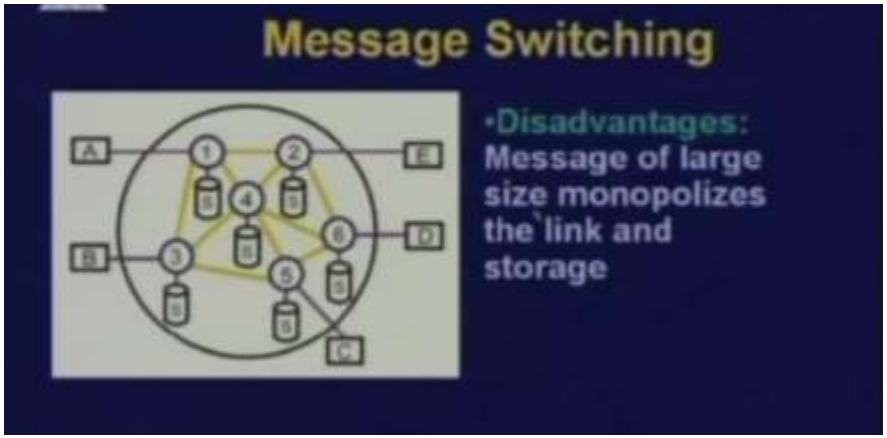
Problems with Circuit Switching

- Network resources are dedicated to a particular connection.
- Two shortcomings for data communication.
  - In a typical user/host data connection, line utilization is very low.
  - Provides facility for data transmission at a constant rate.
    - Data transmission pattern may not ensure this.
    - Limits the utility of the method.

Message Switching

- Basic idea:
  - Each network node receives and stores the message
  - Determines the next leg of the route, and
  - Queues the message to go out on that link.
- Advantages:
  - Line efficiency is greater (sharing of links).
  - Data rate conversion is possible.
  - Even under heavy traffic, packets are accepted, possibly with a greater delivery delay.
  - Message priorities can be used.


Message Switching
From
A
1
2
4
5
To C


Message Switching
A
B
C
D
E
1
2
3
4
5
6
•Disadvantages:
Message of large size monopolizes the`link and storage

Packet Switching

- New form of architecture for long-distance data communication (1970).
- Packet switching technology has evolved over time.
  - basic technology has not changed
  - packet switching remains one of the few effective technologies for long-distance data communication.

Packet Switching: Basic Idea

- Data are transmitted in short packets (few Kbytes).
  - A longer message is broken up into a series of packets.
  - Every packet contains some control information in its header (required for routing and other purposes).

Trailer
CRC
Header
✓ Source address
✓ Destination "
✓ Sequence No

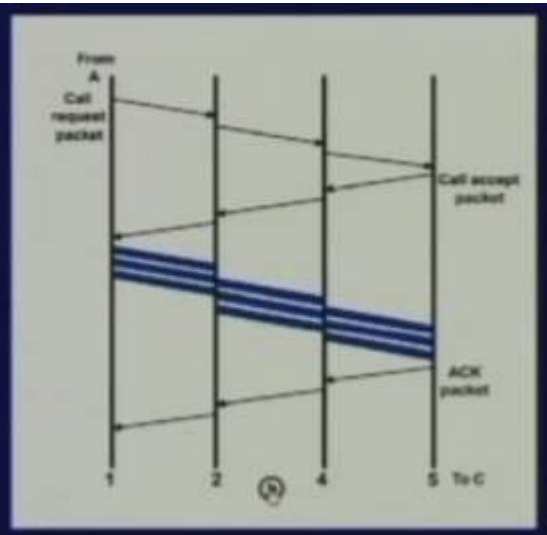
Packet Switching Techniques

- As mentioned earlier, a packet switching network breaks up a message into packets.
- Two approaches are commonly used for handling these packets:
  - Virtual Circuit
  - Datagram

Virtual Circuit Approach

- A preplanned route is established before any packets are sent.
- Call Request and Call Accept packets are used to establish the connection.
- Route is fixed for the duration of the logical connection (like circuit switching).
  - Each packet contains a virtual circuit identifier as well as data.
  - Each node on the route knows where to forward packets.
- A Clear Request packet issued by one of the two stations terminates the connection.

Virtual Circuit

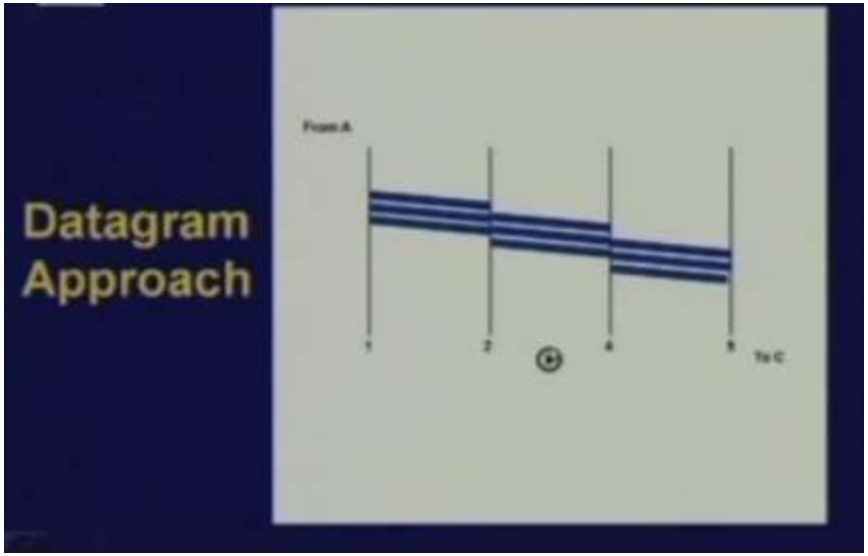


Virtual Circuit

- Main characteristics:
- Route between stations is set up prior to data transfer.
- A packet is buffered at each node, and queued for output over a line.
- A data packet needs to carry only the virtual circuit identifier for effecting routing decisions.
- Intermediate nodes take no routing decisions.
- Often provides sequencing and error control.

Datagram Approach

- Each packet is treated independently, with no reference to packets that have gone before.
- Every intermediate node has to take routing decisions.
- Every packet contains source and destination addresses.
- Intermediate nodes maintain routing tables.



Let us see how the datagram packet switching works. Here as you see (Refer Slide Time: 23:49) a packet is going from 1 to 2 then two other packets are going that is 1 to 2 and the previous packet is going 2 to 4 and three packets are going 1 to 2, 2 to 4 and 4 to 5. So packets of the same messages are going. Also, there is no call request or call forwarding delay so each of these packets can go independently.

Datagram Approach

- Advantages:
  - Call setup phase is avoided (for transmission of a few packets, datagram will be faster).
  - Because it is more primitive, it is more flexible.
  - Congestion/failed link can be avoided (more reliable).
- Problems:
- Packets may be delivered out of order.
- If a node crashes momentarily, all of its queued packets are lost.

Datagram Vs Virtual Circuit

- In virtual circuit — Virtual Circuit Identifier
  - Node need not decide route
  - More difficult to adopt to congestion
  - Maintains sequence order
  - All packets are sent through the same preplanned route
- In datagram service
  - Each packet is treated independently
  - Call set up phase is avoided (Lesser delay)
  - Inherently more flexible and reliable

Comparison

Circuit Switching	Datagram Packet	Virtual Circuit Packet
Dedicated path	No dedicated path	No dedicated path
Path established for entire conversation	Route established for each packet	Route established for entire conversation
Call set up delay	Packet transmission delay	Call set up delay, Packet transmission delay
Overload may block call set up	Overload increases packet delay	Overload may block call set up and increases packet delay
No speed or code conversion	Speed or code conversion	Speed or code conversion
Fixed bandwidth	Dynamic bandwidth	Dynamic bandwidth
No overhead bits after call set up	Overhead bits in each packet	Overhead bits in each packet